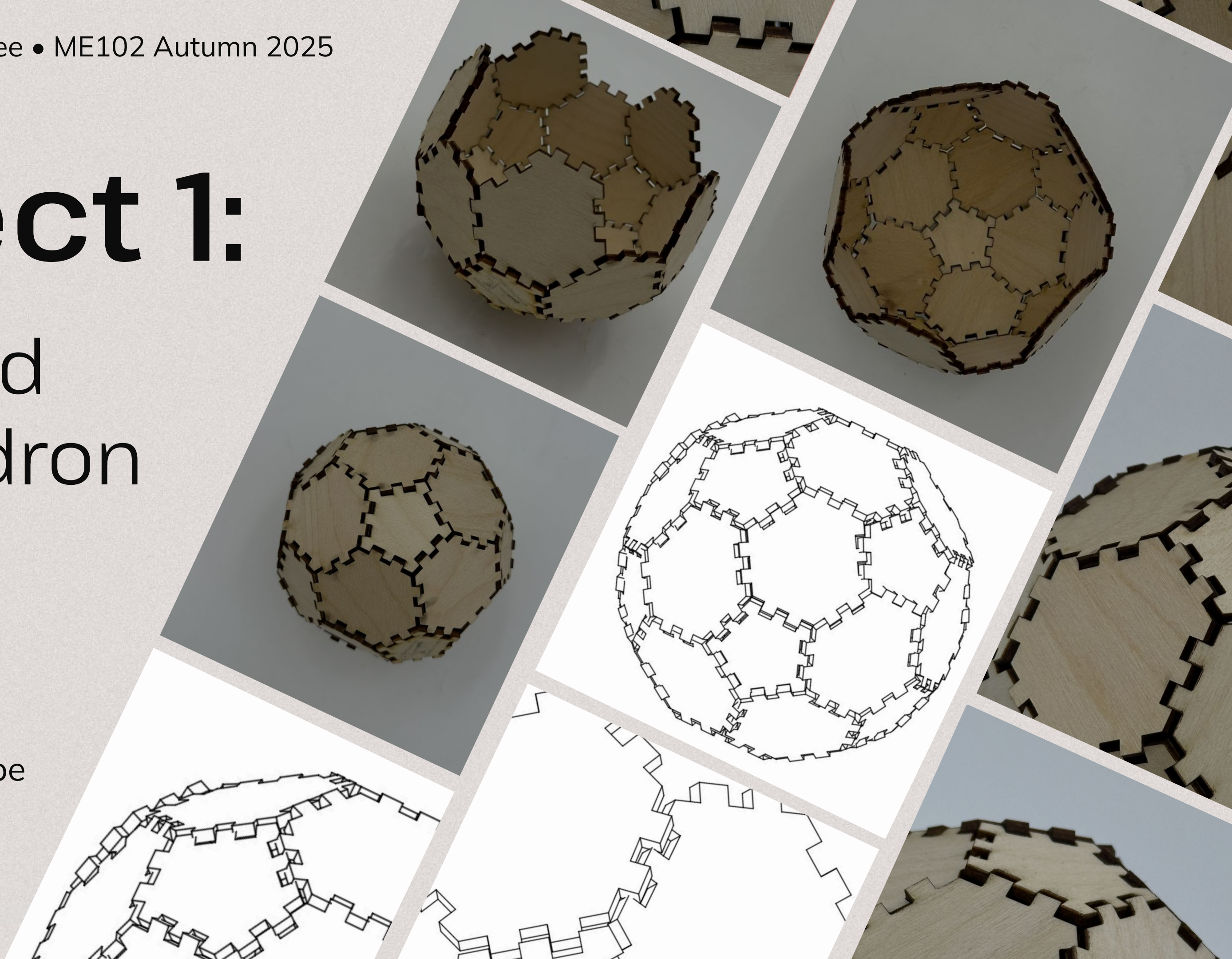
Madison Lee • ME102 Autumn 2025

Project 1:

Truncated Icosahedron

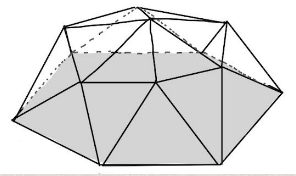
Volume Challenge:
Find the optimal shape
to maximize the ratio

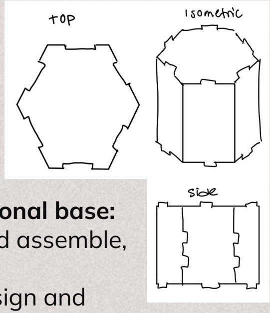


Sketching & Prototyping

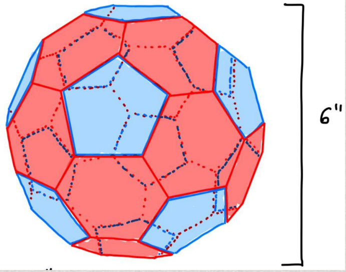
Initial Sketches

Between ease of manufacturing vs. SA-V, I prioritized SA-V. With the most optimal SA-V ratio being a **sphere**, I started by thinking of what were **sphere-adjacent shapes**.





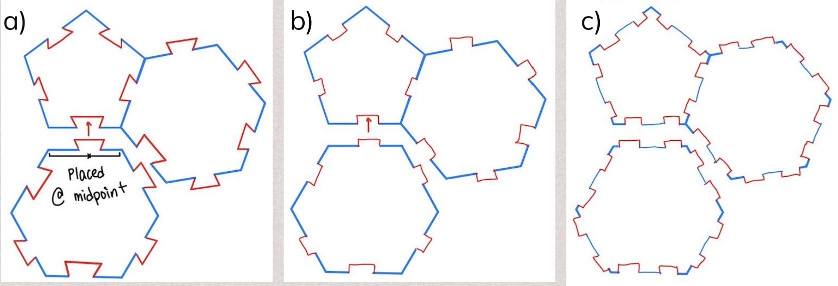
Geodesic dome with a hexagonal base: Cool, but difficult to design and assemble, good SA-V ratio.
Hexagonal Prism: Easy to design and assemble, bad SA-V ratio, standard design
In order to maximize SA-V ratio, had to sacrifice ease in building.



Truncated Icosahedron:

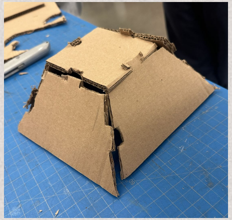
- Maximized SA-V ratio
- Easy to design (2 shapes & symmetric)
- Possible difficult assembly

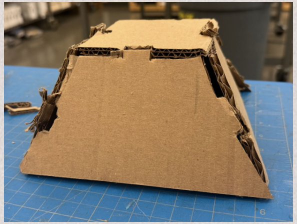
Joints

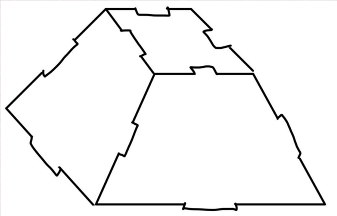


a) **Single Dovetail** – Very secured when interlocked, difficult to do with shape's angle.
b) **Single Friction Fit** – Small surface area, so lower friction, which increases the chances of the object falling apart
c) **Double Friction Fit** – Doubled the surface area, so doubled the friction; ensures interlocking between shapes → more security.

Prototyping

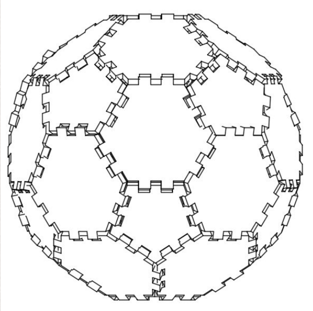






Used the prototype to experiment with joints & found that dovetails work perfectly for 90°, but nothing else. Whereas, friction works for all sorts of angles, if done right.

Final Design

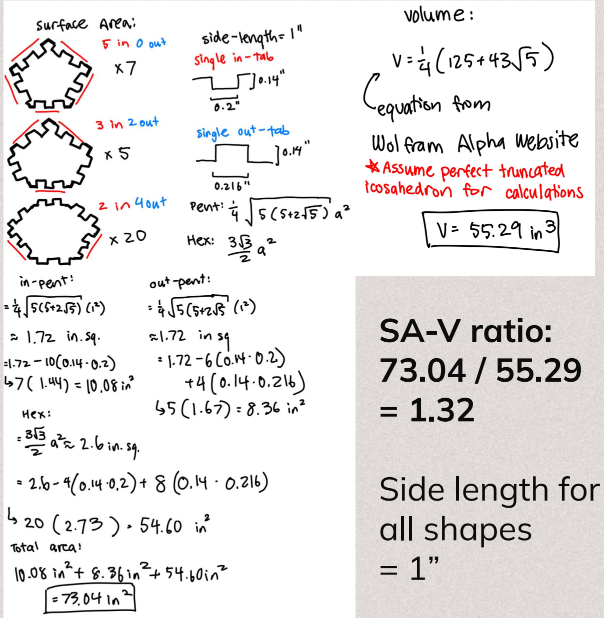


Truncated icosahedron:

- 20 hexagonal pieces
- 12 pentagonal pieces
  - 5 with outer, 7 with only inner
- Double friction fit
- ~ 5.5 inches tall, so 1 inch edges

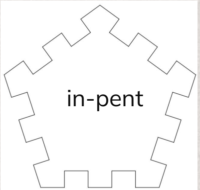
Building & Calculations

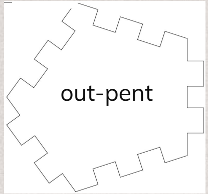
SA-V Ratio

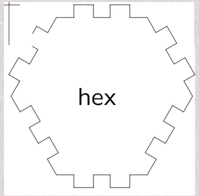


SA-V ratio:
73.04 / 55.29
= 1.32

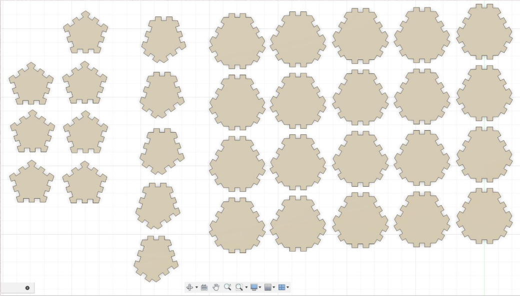
Side length for all shapes
= 1”



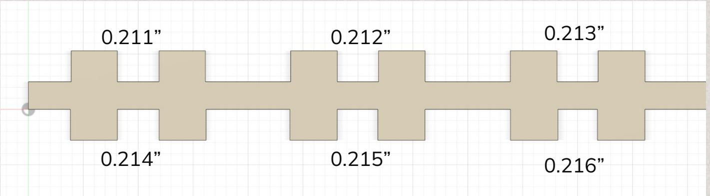




Assembly



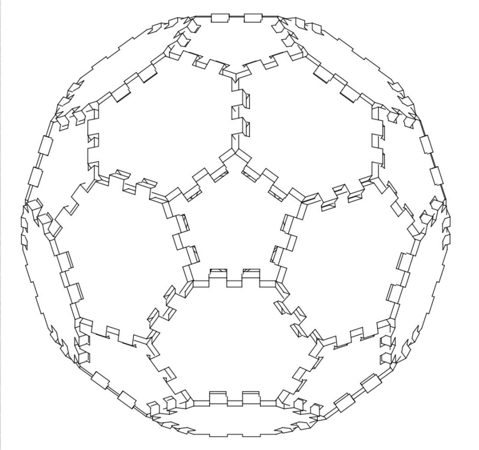
Kerfing & Friction Fit



To determine the tolerances I needed, I cut a variety of insert pieces. The input is constant at 0.2”, and the outer pieces are changing. I initially tried 0.21” width, and the kerf made the two pieces loose enough for them to never stick together. Afterward, I created a stick of extrusions with a variety of widths, from 0.211” to 0.216”. After trial and error, 0.216” was the best fit.

Intrusion: 0.2”
Extrusion: 0.216”

Analysis & Improvements



Analysis

- Designing and prototyping never ends; it's a continual cycle.
    - Sketches allowed me to think every aspect of the project through, from sizing, to connector sizes.
    - Prototypes, especially with a less-structural material, provides valuable insight on manufacturability and challenges the idea at its core of whether or not it is going to work.
- The fabricated components took a lot more trial and error than expected, especially the tolerances.
    - Despite the shapes being symmetrical, the tabs come out different every single time. I didn't know the machines were dependent on which cutter, variability of wood, amount of usage, etc.

Reflection

- I learned the importance of parameters and constraints, and how sketching saves a lot of time in this stage.
- I was heavily debating between having dovetail connectors vs. friction fits until the last minute. However, after creating another cardboard prototype, I realized that unless the pieces were perfectly perpendicular they didn't work, which led me to the friction fit I have now.
- A challenge I ran into was the variability of the cut of the tabs. After the pieces were cut, I noticed that, some of them did not, despite being on the same piece. After realizing that changing the tolerances won't solve the issue, I learned to trust the geometry of the shape, rather than having absolutely perfect connections for every single piece.

Improvements

- I want to find the most ideal overlap for a friction fit. I went to the extent of one hundred thou, but even then, I noticed that some of the parts weren't fitting perfectly.